复旦大学大数据学院

2019 ～ 2020 学年第一学期期末考试试卷

□ A 卷　　☒ B 卷

课程名称：数值算法与案例分析 I

课程代码：DATA130002.01

开课院系：大数据学院 考试形式：闭卷

姓　名： _____ **学　号：** _____ **专　业：** _____

提示：请同学们秉持诚实守信宗旨，谨守考试纪律，摒弃考试作弊。学生如有违反学校考试纪律的行为，学校将按《复旦大学学生纪律处分条例》规定予以严肃处理。

学生（签名）：_____

年　　月　　日

题　目	1	2	3	4	5	6	7	总　分
得　分								

Instructions:

- Write your name and number clearly, and sign the declaration.
- **Write your answers clearly; otherwise, you will lose grades.**
- **Write your answers in detail; otherwise, you will lose grades.**
- Raise your hand if you have any questions.
- Any sort of suspected cheating will result in zero grade of the final.

（装订线内不要答题）

1. (10 pts) Let $\boldsymbol{Q} \in \mathbb{R}^{n\times k}$ $(k < n)$ be an orthonormal matrix (i.e., $\boldsymbol{Q}^\top \boldsymbol{Q} = \boldsymbol{I}$. Compute $\|\boldsymbol{Q}\|_2$ and $\|\boldsymbol{Q}\|_F$.

2. (25 pts) Let $\boldsymbol{u} \in \mathbb{R}^n$ be a unit vector, i.e., $\boldsymbol{u}^\top \boldsymbol{u} = 1$, and let $\boldsymbol{A} = \boldsymbol{u}\boldsymbol{u}^\top$.

   (a) What are the eigenvalues and eigenvectors of $\boldsymbol{A}$?

   (b) Suppose we apply power iteration (starting from an arbitrary eigenvector of unit norm) to compute a single eigenvalue and eigenvector of $\boldsymbol{A}$. Discuss the convergence of the algorithm.

   (c) Suppose we apply power iteration (starting from an arbitrary eigenvector of unit norm) to compute a single eigenvalue and eigenvector of $\boldsymbol{A} - \frac{1}{2}\boldsymbol{I}$, where $\boldsymbol{I} \in \mathbb{R}^{n\times n}$ is an identity matrix. Discuss the convergence of the algorithm.

3. (15 pts) Let

$$\boldsymbol{A} = \begin{bmatrix} -1 & 2 & 2 & 1 \\ 2 & 3 & 0 & -2 \\ 2 & 0 & -1 & 1 \\ 1 & -2 & 1 & -1 \end{bmatrix}.$$

   Compute two orthogonal matrices $\boldsymbol{Q}_1$ and $\boldsymbol{Q}_2$ (for example via Householder transformation) such that $\boldsymbol{Q}_2\boldsymbol{Q}_1\boldsymbol{A}\boldsymbol{Q}_1\boldsymbol{Q}_2$ is a tridiagonal matrix.

4. (10 pts) Let $\boldsymbol{v} \in \mathbb{R}^n$ be a vector of unit norm (i.e., $\|\boldsymbol{v}\|_2 = 1$) and define $\boldsymbol{H} = \boldsymbol{I} - 2\boldsymbol{v}\boldsymbol{v}^\top$, where $\boldsymbol{I} \in \mathbb{R}^{n\times n}$ is an identity matrix. Compute the eigenvalues and eigenvectors of $\boldsymbol{H}$.

5. (10 pts) Let $\boldsymbol{a} \in \mathbb{R}^n \neq 0$ and define

$$\boldsymbol{A} = \underbrace{\begin{bmatrix} \boldsymbol{a} & \cdots & \boldsymbol{a} \end{bmatrix}}_{n \text{ copies of } \boldsymbol{a}} \in \mathbb{R}^{n\times n}.$$

   Compute the SVD of $\boldsymbol{A}$ and find its 2-norm $\|\boldsymbol{A}\|_2$.

6. (15 pts) Consider the following linear equation $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ where

$$\boldsymbol{A} = \begin{bmatrix} 1 & 0 & \alpha \\ 0 & \eta & 0 \\ \gamma & 0 & 1 \end{bmatrix}, \quad \eta \neq 0.$$

   (a) Write out the Jacobi iteration (that is, something like $\boldsymbol{x}_{k+1} = ... \ \boldsymbol{x}_k + ...$) for this linear equation.

   (b) Discuss for what values of α, η and γ Jacobi iteration will converge.

7. (15 pts) Let $\boldsymbol{A}$ be an $m \times n$ matrix of rank n and let $\boldsymbol{b} \in \mathbb{R}^m$. Define

$$f(\boldsymbol{x}) = \frac{1}{2}\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|_2^2 + \frac{\lambda}{2}\|\boldsymbol{x}\|_2^2, \ \lambda > 0.$$

（装订线内不要答题）

In this problem we consider the following optimization problem

$$\min_{\boldsymbol{x}} f(\boldsymbol{x}). \tag{1}$$

Denote by $\boldsymbol{x}_*$ the minimizer of $f(\boldsymbol{x})$. We would like to find $\boldsymbol{x}_*$ by gradient descent method with steepest descent stepsize. Let $\boldsymbol{x}_k$ be the estimate of $\boldsymbol{x}_*$ in the k-th iteration.

(a) Show that $\boldsymbol{x}_*$ is a solution to some linear system (you need to find out the linear system).

(b) Compute the gradient descent direction of $f(\boldsymbol{x})$ at $\boldsymbol{x}_k$, as well as the **steepest descent stepsize** (not backtracking) along the gradient descent direction.

(c) Write out the gradient descent iteration with the steepest descent stepsize for this problem.

（装订线内不要答题）